

**Subject: Civil Engineering Materials (CE 211)****Answer All Questions****Date: 07/11/2021****Time: 11:00 – 12:15 Hrs****Total Marks: 25**

1a) Describe briefly about the differences between ordinary Portland cement and rapid hardening Portland cement. [3]

b) What is the effect of grading of aggregate on workability of concrete mix? [2]

c) Explain about the effect of admixtures on compressive strength of concrete. [4]

d) With a neat diagram, explain about the variation in drying shrinkage of concrete subjected to alternate drying and wetting cycles. [3]

e) Explain about the alkali-aggregate reaction in concrete. [3]

2) Using the following information, design a concrete mix by British mix design method (DOE method) for 'severe' exposure condition:

- Target mean compressive strength of concrete at the age of 28 days = 51.6 N/mm^2 .
- Nominal cover of concrete to steel reinforcement = 30 mm.
- Slump of concrete = 50 mm.
- Cement: ordinary Portland cement (OPC) with specific gravity 3.14.
- The aggregates are of crushed type.
- Maximum size of aggregate = 20 mm, Fine aggregate (sand) is conforming to Grading Zone 3.
- Specific gravity (SSD) of sand = 2.63, Specific gravity (SSD) of coarse aggregate = 2.67.

Calculate the quantities of cement, water, sand, and coarse aggregate for 1 m^3 of the concrete mix. [8]

Also, calculate the revised quantities of the ingredients for 1 m^3 of the concrete mix, if surface moisture of sand, and water absorption of coarse aggregate are 1.5% and 0.7% respectively. [2]

SP : 23 - 1982**TABLE 37 APPROXIMATE COMPRESSIVE STRENGTHS OF CONCRETE MIXES MADE WITH WATER-CEMENT RATIO OF 0.5***(Clause 6.3)*

TYPE OF CEMENT	TYPE OF COARSE AGGREGATE	COMPRESSIVE STRENGTH (N/mm ²)			
		Age (days)			
		3	7	28	91
(1)	(2)	(3)	(4)	(5)	(6)
Ordinary Portland Cement or Sulphate Resisting Portland Cement	Uncrushed	18	27	40	48
	Crushed	23	33	47	55
Rapid Hardening Portland Cement	Uncrushed	25	34	46	53
	Crushed	30	40	53	60

Requirements of BS 8110 : Part 1 : 1985 to ensure durability under specified exposure conditions of reinforced and prestressed concrete made with normal weight aggregate

Condition of exposure	Nominal cover of concrete (mm)				
Mild	25	20	20	20	20
Moderate	-	35	30	25	20
Severe	-	-	40	30	25
Very severe	-	-	50	40	30
Extreme	-	-	-	60	50
Maximum free water/cementitious material ratio	0.65	0.60	0.55	0.50	0.45
Minimum content of cementitious material (kg/m ³)	275	300	325	350	400
Minimum Grade (MPa)	30	35	40	45	50

This table is applicable when maximum size of aggregate is 20 mm.

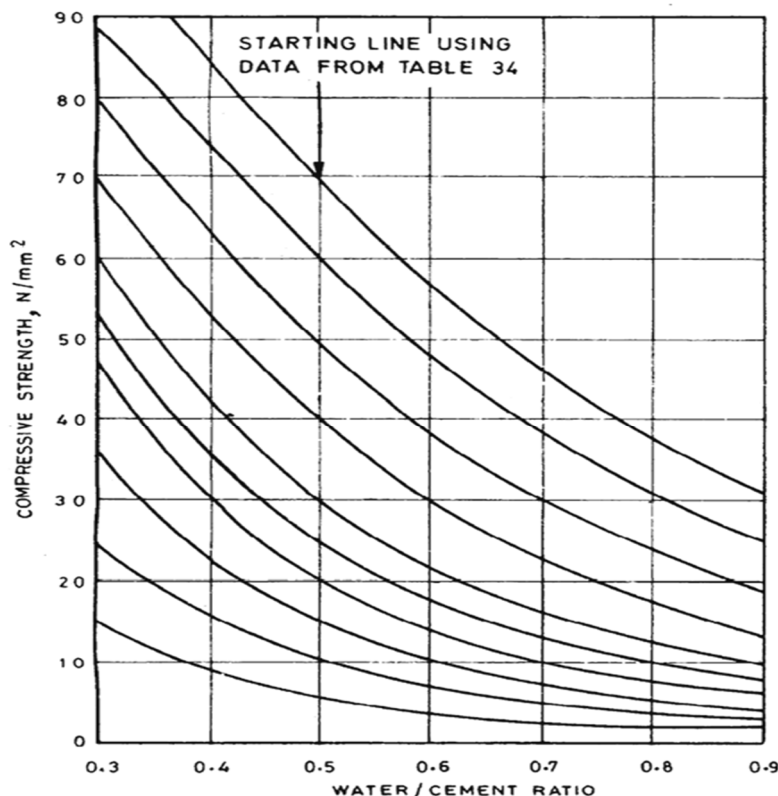
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Fig. 43 Relationship Between Compressive Strength and Water-Cement Ratio

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**TABLE 38 APPROXIMATE WATER CONTENTS
(kg/m³) REQUIRED TO GIVE VARIOUS LEVELS
OF WORKABILITY**

(Clause 6.3)

SLUMP (mm) →		0-10	10-30	30-60	60-180
VEE-BEE (s) →		> 12	6-12	3-6	0-3
MAXIMUM SIZE OF AGGREGATE (mm)	TYPE OF AGGREGATE				
(1)	(2)	(3)	(4)	(5)	(6)
10	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

NOTE 1 — When coarse and fine aggregates of different types are used, the water content is estimated by the expression:

$$\frac{2}{3} W_f + \frac{1}{3} W_c$$

where

W_f = water content appropriate to type of fine aggregate, and

W_c = water content appropriate to type of coarse aggregate.

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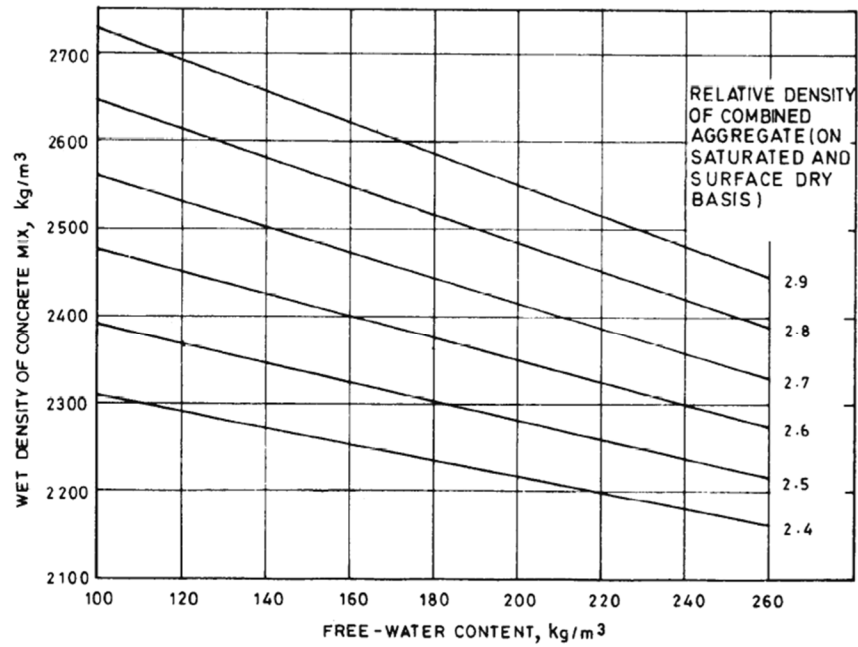


Fig. 44 Estimated Wet Density of Fully Compacted Concrete

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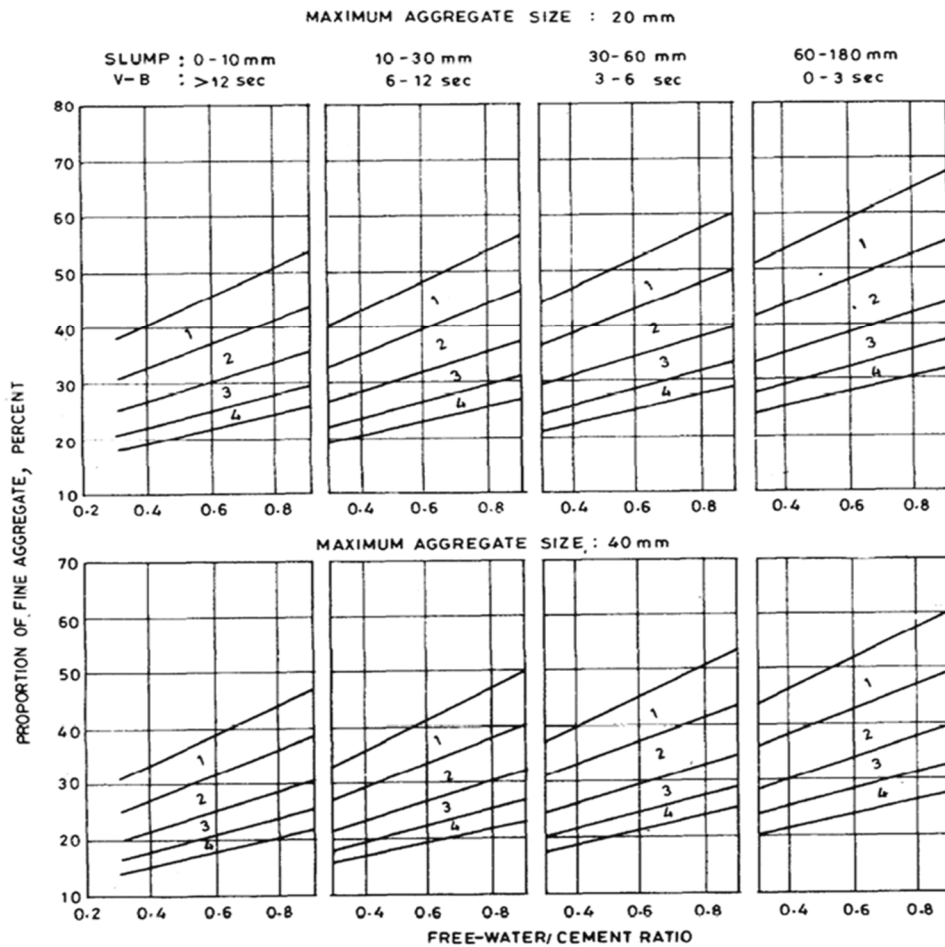


Fig. 45 Recommended Proportions of Fine Aggregate for Grading Zones 1, 2, 3 and 4